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Submitted by

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Sl. No.	Evaluation Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Poor (1)	Marks
1	Problem Definition & Objectives	Problem is clearly defined, relevant, and objectives are specific and measurable	Problem and objectives are clear with minor gaps	Problem is partially defined; objectives need improvement	Problem/objectives are unclear	
2	Technical Approach & Implementation	Appropriate methodology, algorithms/tools used effectively; implementation is complete	Appropriate technical approach with minor issues	Basic implementation with limited technical depth	Incomplete or inappropriate implementation	
3	Innovation / Application / Problem Solving	Demonstrates strong originality, creativity, and effective solution to the identified problem	Shows some innovation and good problem-solving	Limited innovation; mostly conventional approach	No significant innovation/problem-solving	
4	Results, Testing & Analysis	Results are clearly presented, thoroughly tested, and critically analyzed using suitable measures	Results presented with adequate testing and analysis	Basic results and limited testing/analysis	Results are incomplete or unsupported	
5	Documentation, Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation, presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
Total						



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5	Documentation, Presentation & Viva	Well-structured report, clear presentation, excellent communication and confident viva	Good documentation, presentation and satisfactory viva	Adequate documentation/presentation with limited clarity	Poor documentation, presentation or viva	
Total						



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Department of Computer Science & Engineering

DECLARATION

We hereby declare that the entire work embodied in this Mini Project Report titled “Skin Lesion Classification Using Convolutional Neural Networks” has been carried out by us at Sahyadri College of Engineering & Management, Mangaluru under the supervision of Ms. Shiji Abraham, in partial fulfillment of the requirements for the VII semester of Bachelor of Engineering in Computer Science & Engineering. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

Skin lesion classification from dermoscopic images is an important computer vision task in dermatological image analysis. This project presents a deep learning-based approach for classifying dermoscopic images into eight predefined skin lesion categories using convolutional neural networks. The dataset contains 11,720 images and is divided into training, validation, and testing sets using lesion-level grouping to reduce data leakage. A custom CNN was developed as a baseline, followed by transfer learning and fine-tuning using EfficientNetB0. The final fine-tuned model achieved a test accuracy of 75.34%, weighted F1-score of 76.25%, and macro F1-score of 52.70%. Grad-CAM was incorporated to visualize image regions contributing to model predictions and improve model interpretability. A Streamlit-based web application was developed to allow users to upload dermoscopic images and view predicted classes, confidence scores, class probabilities, and Grad-CAM visualizations. The system demonstrates the practical application of deep learning and explainable AI techniques for multi-class skin lesion classification. It is intended for academic and research purposes and is not a substitute for professional medical diagnosis or clinical decision-making.

Table of Contents

Abstract	i
1 Introduction	1
1.1 Overview	1
1.2 Scope & Motivation	1
2 Literature Survey	3
2.1 Recent Deep Learning Approaches for Skin Lesion Classification	3
2.2 EfficientNet and Transfer Learning	3
2.3 Multi-Class CNN and Transformer Methods	4
2.4 Class Imbalance and Data Augmentation	4
2.5 Earlier Transfer-Learning and CNN Studies	5
2.6 Foundational Work	5
2.7 Research Gap	7
3 Problem Formulation	8
3.1 Problem Definition	8
3.2 Objectives	8
3.3 Proposed System	8
3.4 Functional Requirements	9
3.5 Non-Functional Requirements	10
4 Project Design and Implementation	12
4.1 Proposed Project Architecture	12
4.2 High Level Design	13
4.3 Detailed Design	14
4.4 Datasets Descriptions	15
4.4.1 Tools and Technologies Used for Implementation	16
5 Implementation	17
5.1 Experimentation Environment Set Up	17
5.2 Functional Requirements Results Achieved	17
5.3 Non-Functional Requirements Results Achieved	20
5.4 Comparison of Results with Existing Works	21
5.5 Project GUI Snapshots	22
5.6 Societal Impact of the Project	24

5.7	SDG Mapped	24
5.7.1	SDG 3 – Good Health and Well-Being	24
5.7.2	SDG 4 – Quality Education	24
5.7.3	SDG 9 – Industry, Innovation and Infrastructure	24
6	Results and Discussion	26
6.1	Experimental Results	26
6.2	Overall Model Performance	26
6.3	Class-Wise Performance of EfficientNetB0	26
6.4	Comparison of Model Development Stages	27
6.5	Validation and Test Performance	27
6.6	Confusion Matrix Analysis	28
6.7	Grad-CAM Results	28
6.8	Discussion	29
6.9	Summary of Results	29
7	Conclusion and Future Scope	30
7.1	Conclusion	30
7.2	Limitations	30
7.3	Future Scope	31
7.4	Final Summary	31
	References	32

1. Introduction

1.1 Overview

Skin lesions are abnormalities that appear on the skin and may vary considerably in their visual characteristics, making their classification a challenging computer vision problem. Dermoscopic imaging provides detailed visual information about skin lesions and is widely used for analyzing their appearance. However, manual examination of dermoscopic images requires specialized knowledge and can be time-consuming, particularly when dealing with large collections of images.

Deep learning has emerged as an effective approach for image classification because it can automatically learn relevant visual features from image data. Convolutional Neural Networks (CNNs), in particular, are capable of extracting hierarchical features such as edges, textures, shapes, and higher-level patterns from images. Transfer learning further improves this process by using knowledge learned from large image datasets and adapting it to a specific classification task.

This project focuses on developing a deep learning-based system for multi-class skin lesion classification. The system classifies dermoscopic images into eight predefined lesion categories. A custom CNN is first developed as a baseline model, followed by transfer learning and fine-tuning using EfficientNetB0. The final system also incorporates Grad-CAM to visualize image regions contributing to predictions and provides a Streamlit-based interface for practical interaction with the trained model. The project is designed for academic and research purposes and does not replace professional medical diagnosis or clinical decision-making.

1.2 Scope & Motivation

The scope of this project is to develop a deep learning-based system for classifying dermoscopic skin lesion images into eight predefined categories. The project covers the complete workflow from dataset preparation and preprocessing to model training, evaluation, visualization, and deployment through a web-based interface. A custom Convolutional Neural Network (CNN) is implemented as a baseline to establish an initial classification performance. Transfer learning is then applied using EfficientNetB0, followed by fine-tuning to improve the model's ability to learn task-specific visual features.

The project also focuses on improving the interpretability of deep learning predictions. Grad-CAM is incorporated to generate heatmaps that highlight image regions contributing to the model's prediction. This provides a visual explanation of model attention while remaining distinct from clinical reasoning. A Streamlit-based application is developed to allow users to upload dermoscopic images and view the predicted lesion category, confidence score, class probabilities, and Grad-CAM visualization.

The motivation for this project is to explore how modern deep learning techniques can support automated analysis of dermoscopic images. The project also provides an opportunity to study the differences between a custom CNN and a transfer-learning-based architecture and to demonstrate the practical use of explainable AI in image classification. The developed system is intended for educational and research purposes and is not a clinically validated diagnostic system.

2. Literature Survey

2.1 Recent Deep Learning Approaches for Skin Lesion Classification

Erbay et al. presented an ensemble-based framework for multi-class dermoscopic skin lesion classification using CNN and transformer architectures. Their study evaluated architectures including DenseNet201, InceptionResNetV2, and Xception under frozen and fully fine-tuned settings using the HAM10000 dataset. The work demonstrates the current movement toward combining transfer learning, fine-tuning, and ensemble strategies for multi-class lesion classification. This is related to the present project because EfficientNet-based transfer learning and fine-tuning are also used for classification.

Hu et al. proposed a mini-batch sampling and deep metric learning framework to address three important properties of skin-lesion datasets: class imbalance, inter-class similarity, and intra-class variability. Their method specifically recognizes that techniques developed for general image classification may not directly address the characteristics of dermoscopic images. This is relevant to the present work because the project’s dataset also exhibits substantial class imbalance, particularly between common and rare lesion categories.

2.2 EfficientNet and Transfer Learning

Manole et al. investigated EfficientNet for dermatological image classification. Their 2024 study demonstrates the continuing use of EfficientNet architectures in dermatological computer vision and is especially relevant to this project because EfficientNetB0 is used as the final transfer-learning architecture.

Hu et al. proposed a multi-scale feature-fusion network and investigated different approaches for dealing with class imbalance, including class weighting, label smoothing, and resampling. Their experiments used HAM10000 and ISIC 2019, and the authors also examined the importance of lesion and hair-related image features using heat-map analysis. This work is relevant to the present project because it demonstrates the importance of imbalance handling and visual analysis in dermoscopic classification.

Attallah introduced Skin-CAD, an explainable deep-learning system that combines features from multiple CNN architectures and uses LIME for interpretation. The system was evaluated for binary benign/malignant classification and seven-subclass classification. Although the architecture differs from the present EfficientNetB0 implementation, the work directly supports the inclusion of explainability in skin-lesion classification systems.

Wang et al. combined radiomic features, deep learning, and patient metadata for dermoscopic skin-lesion analysis. Their study included datasets and experiments involving different numbers of lesion categories, including an eight-class setting, and investigated both internal and external datasets. The study highlights the potential benefits of combining learned image representations

with additional quantitative information, while also demonstrating that multi-class dermoscopic analysis can involve different class configurations depending on the dataset.

2.3 Multi-Class CNN and Transformer Methods

Raghavendra et al. proposed a deep-learning approach for multi-class skin-lesion classification incorporating global average pooling. Their experiments used the seven-class HAM10000 dataset and compared the proposed CNN with transfer-learning architectures such as ResNet50, VGG16, MobileNetV2, and DenseNet121. The study is relevant to the present project because both works investigate CNN-based multi-class classification and compare customized architectures with transfer-learning models.

Jaisakthi et al. studied several deep neural network architectures for classification of skin cancer from dermoscopic images. Their work reflects the use of deep CNNs to address the fine-grained visual differences between lesion categories. The study provides additional evidence for the usefulness of deep feature learning for dermoscopic classification.

Nie et al. proposed a hybrid CNN-transformer model with focal loss. The CNN component was used to capture local features, while a Vision Transformer model was employed to capture global relationships. Focal loss was used to address the extreme class imbalance present in the ISIC 2018 dataset. This work shows how newer architectures attempt to combine local convolutional features with global attention mechanisms.

2.4 Class Imbalance and Data Augmentation

Qian et al. introduced grouping of multi-scale attention blocks together with class-specific loss weighting for skin-lesion classification. Their objective was to improve fine-grained feature extraction while addressing class imbalance. The method was evaluated on HAM10000 and achieved 91.6% accuracy and 97.1% AUC. The study is particularly relevant because the present project also deals with a highly imbalanced multi-class problem and evaluates individual class precision, recall, and F1-score rather than relying only on accuracy.

Shen et al. developed a low-cost data-augmentation strategy using EfficientNets and evaluated it on HAM10000, ISIC 2017, and Derm7pt. Their work showed that augmentation can improve classification performance while controlling computational cost. They also used Grad-CAM++ heatmaps to examine regions associated with model decisions. This is closely related to the present project because it combines EfficientNet-based learning, data augmentation, and model visualization.

Wu et al. conducted a systematic review of deep-learning methods for skin cancer classification. The review identified several recurring challenges, including data imbalance, limited training data, domain adaptation, model robustness, and model efficiency. It also summarized CNN-based approaches, data augmentation, and other strategies used to improve skin-lesion classification. Benyahia et al. investigated deep CNN feature extraction using pretrained architectures together with multiple machine-learning classifiers. They evaluated the approach using the ISIC 2019

and PH2 datasets and found that pretrained CNN features can provide effective representations for lesion classification. The work demonstrates the usefulness of transfer learning even when the final classifier is different from the original neural-network classification head.

Popescu et al. investigated collective intelligence by combining predictions from multiple CNN architectures, including AlexNet, MobileNet-V2, Xception, ResNet, InceptionResNet-V2, and DenseNet201. Their work used HAM10000 and showed that combining multiple networks could improve validation performance compared with individual networks. This provides a useful comparison with the present project, which deliberately uses a single EfficientNetB0 model rather than an ensemble.

2.5 Earlier Transfer-Learning and CNN Studies

Calderón et al. proposed BILSK, a bilinear CNN approach that combined ResNet50 and VGG16 for seven-class HAM10000 classification. Data augmentation, transfer learning, and fine-tuning were used, and the proposed bilinear model achieved an accuracy of 0.9321. This study demonstrates how combining CNN representations can improve lesion classification and provides a useful contrast with the single-backbone EfficientNetB0 approach used in the present project.

Srinivasu et al. proposed a MobileNetV2-LSTM based skin-disease classification system using HAM10000. The work emphasized computational efficiency and also developed a mobile application for image-based classification. It is relevant to the present project because both works consider practical deployment of deep-learning skin-image classifiers rather than focusing only on offline experimentation.

Gessert et al. developed an ensemble of multi-resolution EfficientNets for the ISIC 2019 Skin Lesion Classification Challenge. The authors addressed severe class imbalance using loss balancing and used multiple image resolutions and cropping strategies. Their study is particularly relevant because EfficientNet is also the core transfer-learning family used in the present project. Mahbod et al. investigated transfer learning using multiple fine-tuned CNNs and multi-scale cropped images for skin-lesion classification. Their experiments demonstrated the usefulness of transfer learning and multi-scale processing for dermoscopic images. The work supports the use of pretrained networks as a practical alternative to training deep architectures entirely from scratch.

Hosny et al. applied transfer learning using AlexNet to classify seven categories of skin lesions. Their study demonstrates an early application of pretrained CNNs to multi-class dermoscopic classification and provides a useful baseline for understanding the progression from earlier architectures toward newer networks such as EfficientNet.

2.6 Foundational Work

Tan and Le introduced EfficientNet through a compound scaling strategy that jointly scales network depth, width, and input resolution. Their work demonstrated that carefully balanced model

scaling can achieve strong accuracy with improved computational efficiency. EfficientNetB0 used in the present project is part of this architecture family.

Tschandl, Rosendahl, and Kittler introduced the HAM10000 dataset, containing 10,015 dermatoscopic images collected from multiple sources. The dataset became an important benchmark for skin-lesion classification and has been used extensively in later deep-learning research. It is important to distinguish this original seven-category benchmark from the eight-class dataset configuration used in the present project.

Selvaraju et al. introduced Grad-CAM, a gradient-based visualization technique that highlights image regions contributing to a model’s class prediction. Grad-CAM is particularly useful for improving the interpretability of convolutional neural networks. In the present project, the technique is used to generate heatmaps for EfficientNetB0 predictions. Importantly, such heatmaps represent model-related visual importance and should not be interpreted as independent clinical explanations.

Table 2.1: Comparative Summary of Related Literature

Year	Authors	Main Approach	Dataset / Setting	Relevance to Present Project
2025	Erbay et al.	CNN and transformer ensemble with transfer learning and fine-tuning	HAM10000	Related to multi-class classification, transfer learning, fine-tuning, and modern deep-learning architectures
2025	Hu et al.	Mini-batch sampling and deep metric learning	Skin-lesion classification	Addresses class imbalance, inter-class similarity, and intra-class variability
2024	Wang et al.	Radiomics, deep learning, and patient metadata	Multiple dermatoscopic datasets; different class configurations	Shows the potential of combining image features with additional information in multi-class lesion analysis
2024	Attallah	Skin-CAD using dual CNN features and transfer learning	Binary and seven-subclass classification	Relevant to transfer learning and explainable skin-lesion classification
2024	Hu et al.	Multi-scale feature fusion and class-weighted learning	HAM10000 and ISIC 2019	Addresses class imbalance and visual feature analysis
2024	Manole et al.	EfficientNet-based deep learning	Dermatological image classification	Directly relevant to the use of EfficientNet in the present project
2023	Raghavendra et al.	CNN with global average pooling and transfer learning	HAM10000	Related to CNN-based multi-class classification and transfer-learning comparison
2023	Jaisakthi et al.	Deep neural network architectures	Dermoscopic images	Demonstrates the application of deep CNNs to skin lesion classification
2023	Nie et al.	CNN–Vision Transformer hybrid with focal loss	ISIC 2018	Combines local CNN features and global attention while addressing class imbalance

Table 2.1 summarizes the major studies reviewed in this literature survey, covering recent CNN, transformer, EfficientNet, transfer-learning, ensemble, class-imbalance, and explainability approaches. The reviewed studies demonstrate the progression of skin-lesion classification from conventional CNN and transfer-learning methods toward more advanced architectures and hybrid

approaches. The table also highlights recurring research challenges such as class imbalance, inter-class similarity, and model interpretability.

2.7 Research Gap

The reviewed literature shows a progression from conventional CNNs and early transfer-learning approaches toward EfficientNet, multi-scale networks, attention mechanisms, CNN-transformer hybrids, ensembles, and explainable AI. At the same time, class imbalance, inter-class similarity, intra-class variation, computational requirements, and model interpretability remain recurring challenges.

Most benchmark studies use datasets such as HAM10000 or ISIC with specific class configurations, commonly seven classes. Therefore, published performance values cannot be directly equated with the results of the present project, which uses its own 8-class, 11,720-image dataset and experimental split. The present project addresses the practical end-to-end workflow by combining a custom CNN baseline, EfficientNetB0 transfer learning and fine-tuning, multi-class evaluation, Grad-CAM visualization, and a Streamlit interface.

3. Problem Formulation

3.1 Problem Definition

The classification of skin lesions from dermoscopic images is a challenging multi-class image classification problem because different lesion categories can exhibit similar visual characteristics, while images within the same category may show considerable variation in color, texture, shape, and appearance. Manual analysis of a large number of dermoscopic images can also be time-consuming and requires specialized expertise.

The problem addressed in this project is to develop a deep learning-based system capable of classifying dermoscopic skin lesion images into eight predefined categories. The system should learn meaningful visual patterns from the available dataset and provide reliable predictions for previously unseen images. Since the dataset contains an unequal number of samples across the different lesion categories, evaluation should consider class-wise performance rather than relying only on overall accuracy.

The project also addresses the interpretability of deep-learning predictions. A classification model may provide a predicted class and confidence value without indicating which parts of the image contributed to that prediction. Therefore, Grad-CAM is incorporated to visualize regions of the input image that contribute to the model's predicted class.

The complete system includes dataset preparation, image preprocessing, CNN-based baseline development, transfer learning using EfficientNetB0, fine-tuning, performance evaluation, Grad-CAM visualization, and deployment through a Streamlit-based web application.

3.2 Objectives

The main objectives of the proposed system are:

1. To develop a deep learning-based system for classifying dermoscopic images into eight predefined skin lesion categories.
2. To develop and compare a custom CNN model with an EfficientNetB0-based transfer learning and fine-tuning approach for skin lesion classification.
3. To evaluate the performance of the developed models using accuracy, precision, recall, F1-score, confusion matrix, and class-wise performance.
4. To improve model interpretability using Grad-CAM and integrate the final model into a Streamlit-based web application for image prediction and visualization.

3.3 Proposed System

The proposed system follows an end-to-end deep learning pipeline for skin lesion classification. Initially, dermoscopic images are collected and organized according to their corresponding lesion

categories. The dataset is divided into training, validation, and testing subsets using lesion-level grouping to reduce the possibility of related images appearing across different subsets.

The images are then preprocessed and supplied to the deep learning models. A custom CNN is trained first to establish a baseline. Subsequently, EfficientNetB0 with pretrained weights is used as a transfer-learning model. The classification head is adapted to the eight target classes, and selected layers are fine-tuned using the project dataset.

After training, the models are evaluated using the validation and test datasets. Accuracy, precision, recall, F1-score, and confusion matrices are used to analyze overall and class-specific performance. The final EfficientNetB0 model is selected for inference based on the experimental results.

For interpretability, Grad-CAM is applied to the trained model to generate heatmaps showing regions that contribute to a particular prediction. Finally, the trained model is integrated into a Streamlit web application. The application accepts a dermoscopic image as input and displays the predicted lesion category, confidence score, class probabilities, and corresponding Grad-CAM visualization.

The system is intended as an academic and research prototype and should not be considered a clinically validated diagnostic system.

3.4 Functional Requirements

The functional requirements describe the core operations and services provided by the proposed skin lesion classification system. The system is designed to accept dermoscopic images as input, perform automated classification using the trained deep learning model, and present the prediction results together with an interpretability visualization. The major functional requirements of the proposed system are as follows.

FR1. Image Input

The system should provide an interface for uploading a dermoscopic skin lesion image for classification. The uploaded image is treated as the input to the prediction pipeline.

FR2. Image Preprocessing

The system should preprocess the uploaded image according to the input requirements of the trained classification model before performing inference.

FR3. Multi-Class Classification

The system should classify the input image into one of the eight predefined skin lesion categories: `akiec`, `bcc`, `bkl`, `df`, `mel`, `nv`, `scc`, and `vasc`.

FR4. Prediction Probability

The system should generate and display the predicted class along with its corresponding confidence score and probability distribution across all eight classes.

FR5. Model Interpretability

The system should generate a Grad-CAM visualization for the model prediction. The resulting heatmap should indicate the image regions contributing to the model's classification.

FR6. Model Evaluation

The system should support quantitative evaluation of the trained classification models using accuracy, precision, recall, F1-score, and confusion matrix analysis.

FR7. Result Visualization

The system should present classification results and Grad-CAM visualizations through a web-based interface to facilitate interpretation of the model output.

3.5 Non-Functional Requirements

The non-functional requirements specify the quality attributes and operational characteristics of the proposed system. These requirements focus on the usability, reliability, maintainability, performance, and appropriate operation of the developed application.

NFR1. Performance

The system should perform image inference and generate classification results within a reasonable response time after an image is uploaded.

NFR2. Usability

The application should provide a simple and intuitive interface through which users can upload an image and access the classification results without requiring knowledge of the underlying implementation.

NFR3. Reliability

The system should produce consistent outputs when the same valid input image is processed using the same model and configuration.

NFR4. Maintainability

The implementation should be organized into modular components for data processing, model inference, evaluation, Grad-CAM generation, and application development, allowing individual components to be modified without substantially affecting the complete system.

NFR5. Scalability

The system architecture should allow the trained classification model to be replaced or updated with improved architectures without requiring major changes to the application interface.

NFR6. Interpretability

The system should provide visual explanations through Grad-CAM in addition to the predicted class and confidence values, thereby providing supplementary information about the regions influencing model predictions.

NFR7. Portability

The application should be capable of execution in environments supporting the required Python, TensorFlow/Keras, and Streamlit dependencies.

NFR8. Safety and Appropriate Use

The system should communicate that the generated predictions are intended for academic and research purposes and should not be interpreted as a substitute for professional medical diagnosis or clinical decision-making.

4. Project Design and Implementation

4.1 Proposed Project Architecture

The proposed system is designed as an end-to-end deep learning pipeline for multi-class skin lesion classification from dermoscopic images. The architecture consists of several interconnected stages, beginning with dataset preparation and ending with prediction visualization through a web-based interface.

A custom Convolutional Neural Network (CNN) is implemented as the baseline architecture. The baseline results are subsequently used for comparison with a transfer-learning approach based on EfficientNetB0. The pretrained EfficientNetB0 model is adapted to the eight-class classification problem and subsequently fine-tuned using the project dataset.

The trained model generates a probability distribution over the eight lesion categories. The class with the highest predicted probability is selected as the model prediction.

To provide an additional view of the model's prediction behavior, Grad-CAM is applied to the top convolutional layer of EfficientNetB0. The resulting visualization identifies the image regions contributing to the prediction and provides a visual representation of the model's attention.

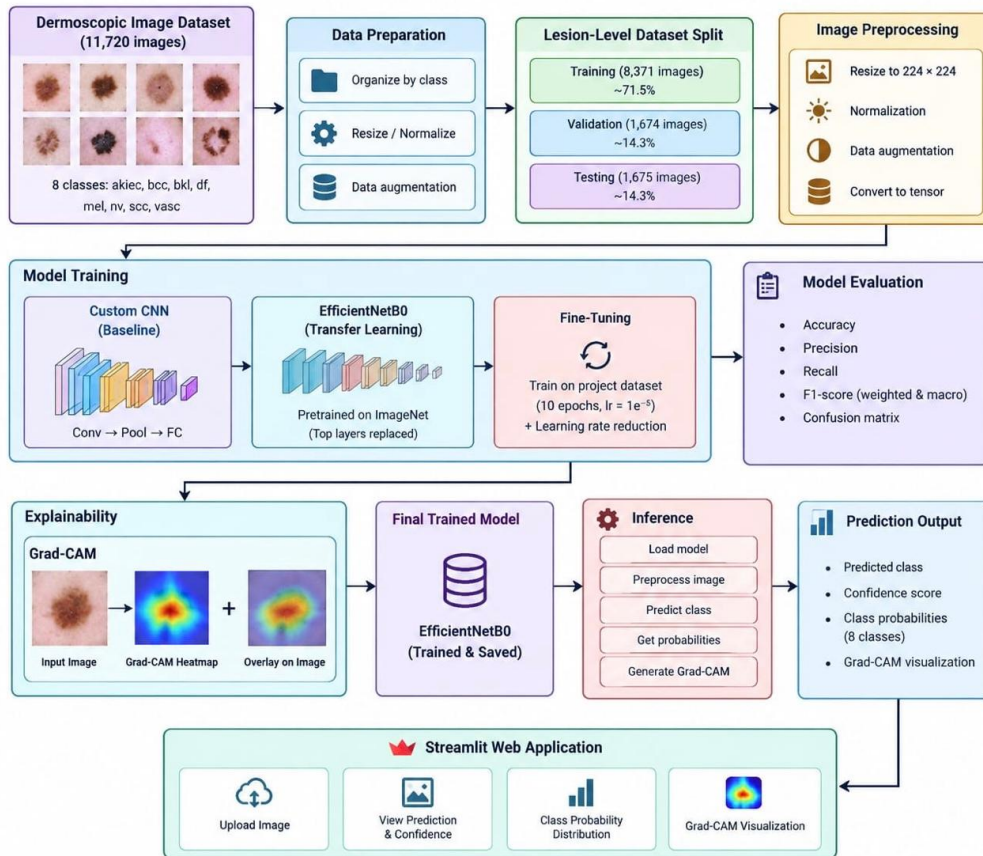


Figure 4.1: Proposed Project Architecture

4.2 High Level Design

The high-level design of the proposed system is divided into a sequence of clearly defined processing stages. Each stage performs a specific function and passes its output to the next stage. This modular structure makes the complete workflow easier to understand, implement, and maintain.

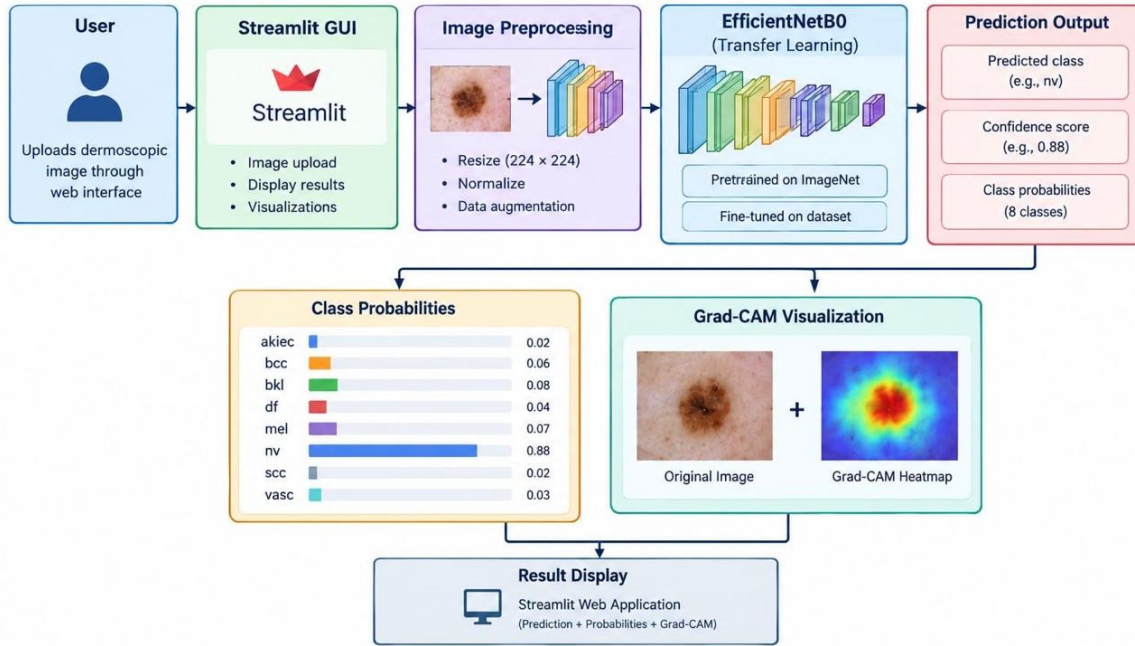


Figure 4.2: High Level Design

Input Layer

The input layer accepts a dermoscopic skin lesion image from the user through the web-based application. The uploaded image acts as the primary input to the complete prediction pipeline.

Image Preprocessing Layer

The uploaded image is preprocessed according to the input requirements of the trained deep learning model. The preprocessing stage prepares the image in the required size and numerical format before it is supplied to the model.

Feature Extraction and Classification Layer

The preprocessed image is passed to the trained EfficientNetB0 model. The network extracts hierarchical visual features from the image and uses the learned representation for classification into one of the eight predefined skin lesion categories.

Prediction Layer

The classification layer produces a probability distribution across the eight target classes. The class with the highest predicted probability is selected as the final model prediction. The system also retains the probability values for all eight classes for further interpretation.

Grad-CAM Explainability Layer

Grad-CAM is applied to the trained EfficientNetB0 model to generate a heatmap highlighting the image regions that contribute to the predicted class. This provides a visual indication of model attention and supports interpretation of the prediction.

Output and Presentation Layer

The final prediction is presented through the Streamlit web application. The interface displays the predicted lesion category, confidence score, class probabilities, and the corresponding Grad-CAM visualization to the user.

4.3 Detailed Design

The detailed design describes the internal processing stages of the proposed system. The input dermoscopic image is first resized and normalized before being provided to the trained deep learning model.

The EfficientNetB0 architecture extracts hierarchical visual features from the input image through a sequence of convolutional and feature-extraction operations. The final classification layer produces probability values corresponding to the eight target classes. The class having the highest probability is selected as the predicted lesion category.

For interpretability, Grad-CAM uses the gradients associated with the predicted class to generate a heatmap over the input image. The heatmap indicates the regions that have a stronger influence on the model's prediction.

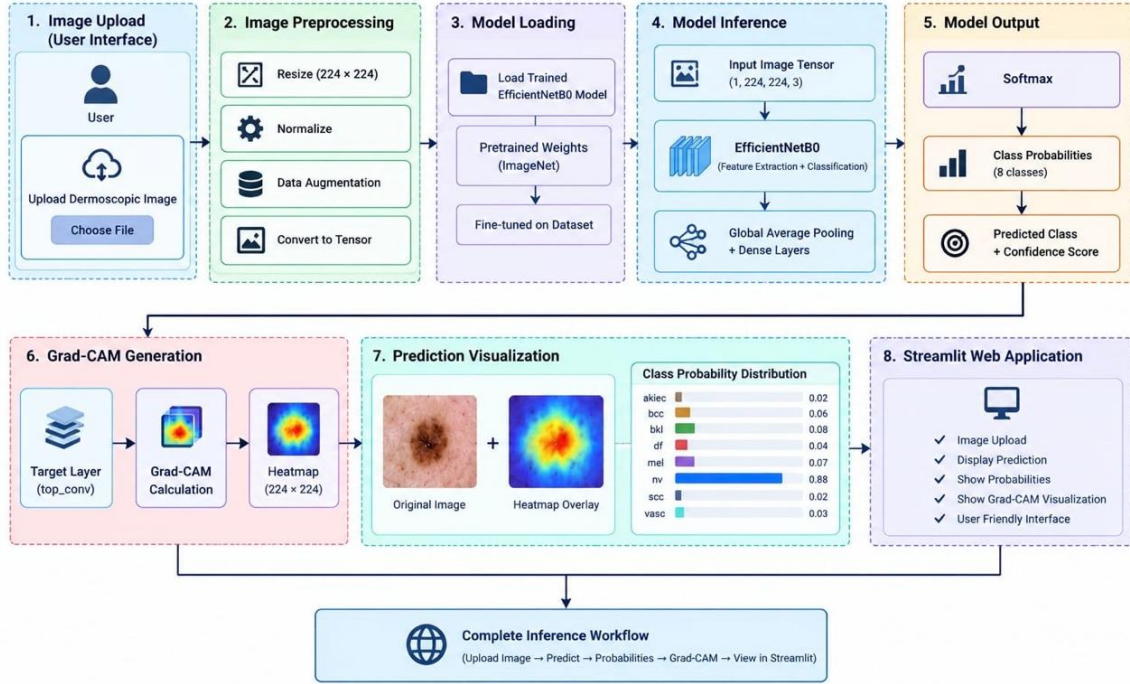


Figure 4.3: Detailed Design of the Proposed System

4.4 Datasets Descriptions

The dataset used in the project consists of **11,720 dermoscopic images** belonging to eight predefined lesion categories. The dataset exhibits substantial class imbalance, with **nv** containing considerably more images than several of the minority classes.

The dataset is divided into training, validation, and testing subsets. The validation set contains **1,674 images**, while the test set contains **1,675 images**. The remaining **8,371 images** constitute the training set.

Table 4.1: Dataset Split

Dataset Subset	Number of Images
Training	8,371
Validation	1,674
Testing	1,675
Total	11,720

Table 4.2: Target Lesion Categories

Class Code	Lesion Category
akiec	Actinic keratosis
bcc	Basal cell carcinoma
bk1	Benign keratosis-like lesion
df	Dermatofibroma
mel	Melanoma
nv	Melanocytic nevus
scc	Squamous cell carcinoma
vasc	Vascular lesion

The unequal distribution of samples among these categories is an important consideration during model evaluation. Consequently, class-wise precision, recall, and F1-score are reported in addition to overall accuracy.

4.4.1 Tools and Technologies Used for Implementation

The project was implemented using Python-based machine learning and web-development technologies.

Table 4.3: Tools and Technologies

Tool / Technology	Purpose
Python	Main programming language
TensorFlow / Keras	CNN development, transfer learning, training and inference
EfficientNetB0	Transfer-learning and fine-tuning architecture
NumPy	Numerical computation
Pandas	Data handling and result processing
Scikit-learn	Performance evaluation and classification metrics
Matplotlib	Graph and result visualization
Grad-CAM	Model interpretability
Streamlit	Web application development
Git / GitHub	Source-code version control and project repository
NVIDIA GeForce RTX 3050 Laptop GPU	GPU-accelerated model training

5. Implementation

5.1 Experimentation Environment Set Up

The experiments were conducted using a Python-based development environment with TensorFlow/Keras as the primary deep learning framework. GPU acceleration was enabled using an **NVIDIA GeForce RTX 3050 Laptop GPU**.

The installed TensorFlow version was **2.21.0**. Supporting libraries included NumPy, Pandas, Scikit-learn, and Matplotlib. The trained EfficientNetB0 model was saved in Keras format and subsequently used for evaluation, inference, Grad-CAM generation, and Streamlit deployment. The experimentation consisted of three principal model stages: the original custom CNN, the stabilized custom CNN, and the final fine-tuned EfficientNetB0 model.

5.2 Functional Requirements Results Achieved

The functional requirements defined in Section 3.4 were implemented and demonstrated through the developed system.

Table 5.1: Functional Requirements Achievement

Requirement	Implementation / Result
Image input	Dermoscopic image upload is supported through the Streamlit interface
Image preprocessing	Uploaded images are processed before model inference
Multi-class classification	Images are classified into eight predefined lesion categories
Prediction probability	Predicted class, confidence, and class probabilities are displayed
Model interpretability	Grad-CAM heatmaps are generated using EfficientNetB0
Model evaluation	Accuracy, precision, recall, F1-score, and confusion matrices are generated
Result visualization	Prediction and Grad-CAM outputs are presented through the web interface

The developed system therefore provides the complete functional workflow from image input to classification and visualization.

Table 5.2: Overall Model Performance

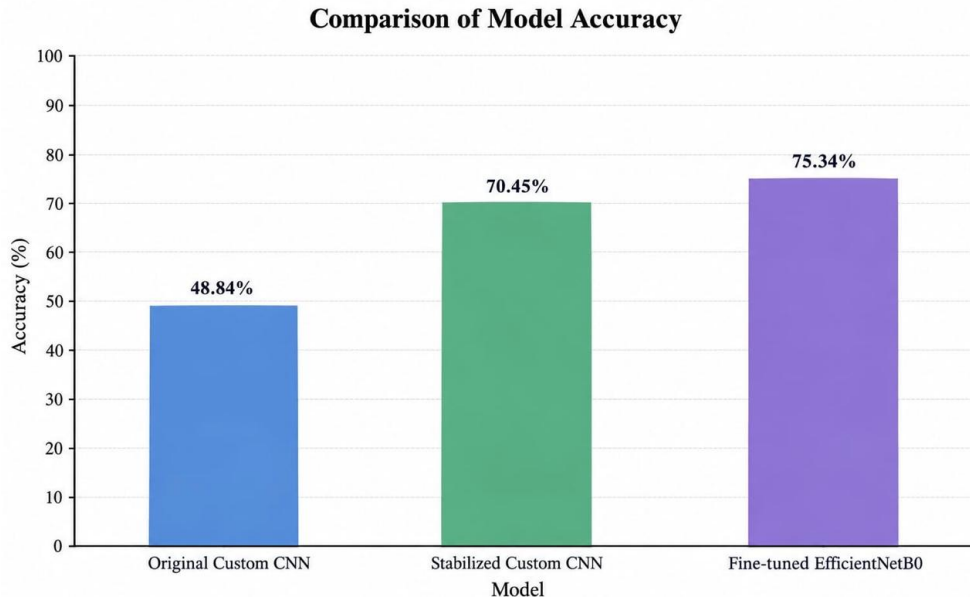
Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score	Macro F1-Score
Original Custom CNN	48.84%	69.74%	48.84%	53.20%	—
Stabilized Custom CNN	70.45%	69.44%	70.45%	69.65%	41.59%
Fine-tuned EfficientNetB0	75.34%	77.66%	75.34%	76.25%	52.70%

Result Discussion

The original custom CNN obtained a test accuracy of **48.84%**. Following stabilization, the custom CNN achieved **70.45%**. The fine-tuned EfficientNetB0 produced the highest measured overall accuracy at **75.34%**.

The weighted F1-score increased from **53.20%** for the original CNN to **69.65%** for the stabilized CNN and **76.25%** for EfficientNetB0. The increase indicates improved overall classification performance under the weighted evaluation.

However, the macro F1-score remained lower than the weighted F1-score, indicating that model performance was not uniform across all lesion categories.

**Figure 5.1:** Comparison of Model Accuracy

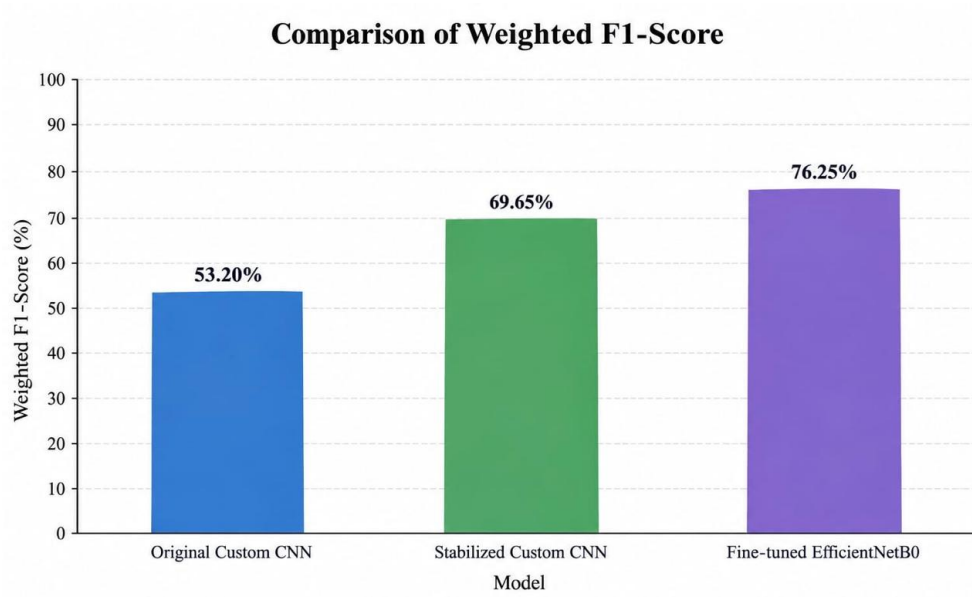


Figure 5.2: Comparison of Weighted F1-Score

Table 5.3: Class-Wise Performance of Fine-Tuned EfficientNetB0

Class	Precision	Recall	F1-Score	Support
akiec	0.38	0.36	0.37	22
bcc	0.59	0.55	0.57	89
bkl	0.50	0.63	0.56	191
df	0.33	0.59	0.42	22
mel	0.51	0.59	0.55	186
nv	0.92	0.85	0.88	1106
scc	0.21	0.21	0.21	33
vasc	0.65	0.65	0.65	26

The class-wise results demonstrate considerable variation in classification performance. The **nv** category achieves the highest F1-score of **0.88**, whereas **scc** obtains the lowest F1-score of **0.21**. The results also illustrate the effect of class imbalance, since the **nv** category has substantially greater representation in the test set.

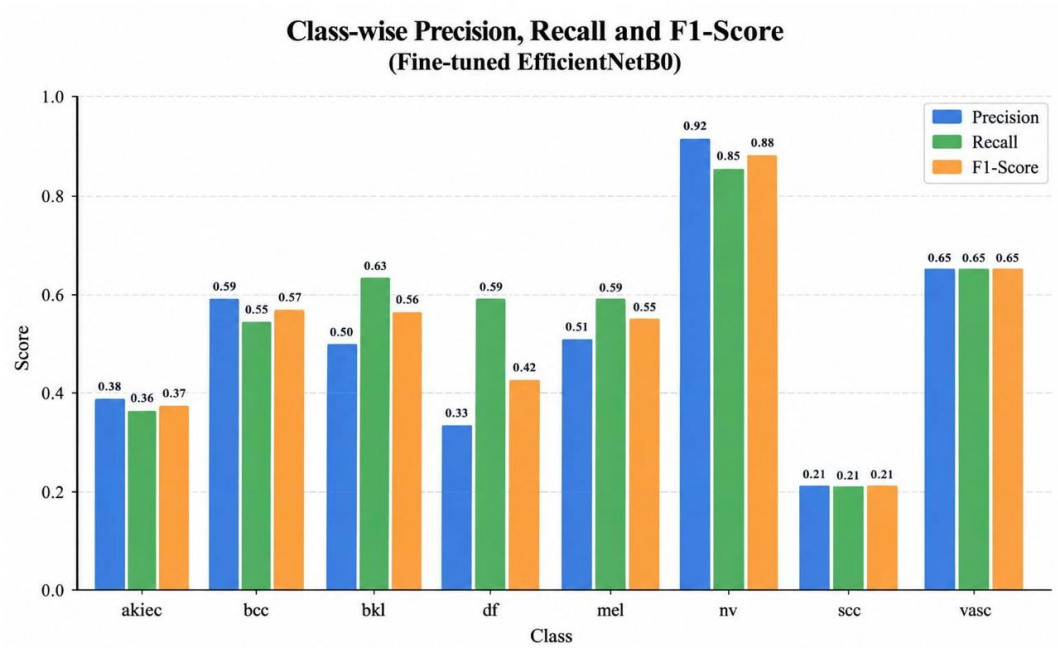


Figure 5.3: Class-Wise F1-Score

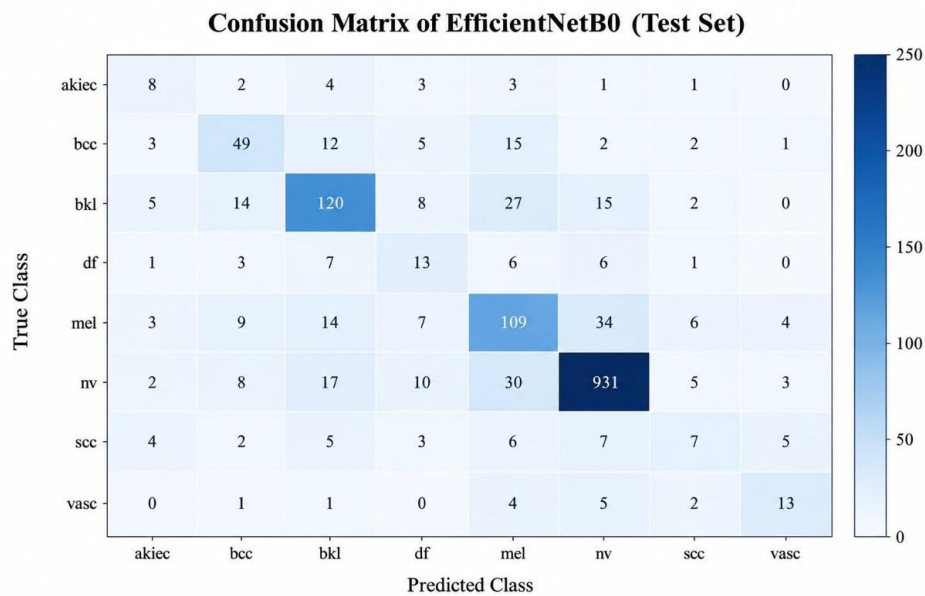


Figure 5.4: Confusion Matrix of EfficientNetB0

The confusion matrix provides a class-by-class representation of correct and incorrect predictions and should be discussed alongside the class-wise performance table. The project-generated confusion matrix file is `results/efficientnet_b0_finetuned_confusion_matrix.csv`.

5.3 Non-Functional Requirements Results Achieved

The non-functional requirements were assessed primarily through implementation behavior and application usability.

Table 5.4: Non-Functional Requirements Achievement

Requirement	Observed Outcome
Performance	GPU acceleration was used during model training; inference is integrated into the application workflow
Usability	A Streamlit interface provides image upload and result visualization
Reliability	The saved trained model and fixed inference pipeline allow consistent processing of the same input under the same configuration
Maintainability	Model, inference, evaluation, and Grad-CAM functionality are implemented as separate components
Scalability	The application can accommodate future model improvements through the model/inference pipeline
Interpretability	Grad-CAM visualization is provided alongside predictions
Portability	The application can run in environments containing the required Python dependencies
Safety and appropriate use	The application is positioned as an academic/research prototype rather than a clinical diagnostic system

For the non-functional requirements, quantitative measurements such as exact inference latency, memory consumption, or concurrent-user capacity were not part of the recorded experiments. Therefore, these characteristics should be presented as **implementation-level observations rather than numerical performance claims**.

5.4 Comparison of Results with Existing Works

The performance of the proposed model should be compared with previous studies reviewed in Chapter 2. However, the reported values cannot be interpreted as a direct model-to-model ranking because the studies use different datasets, class configurations, preprocessing strategies, experimental protocols, and evaluation procedures.

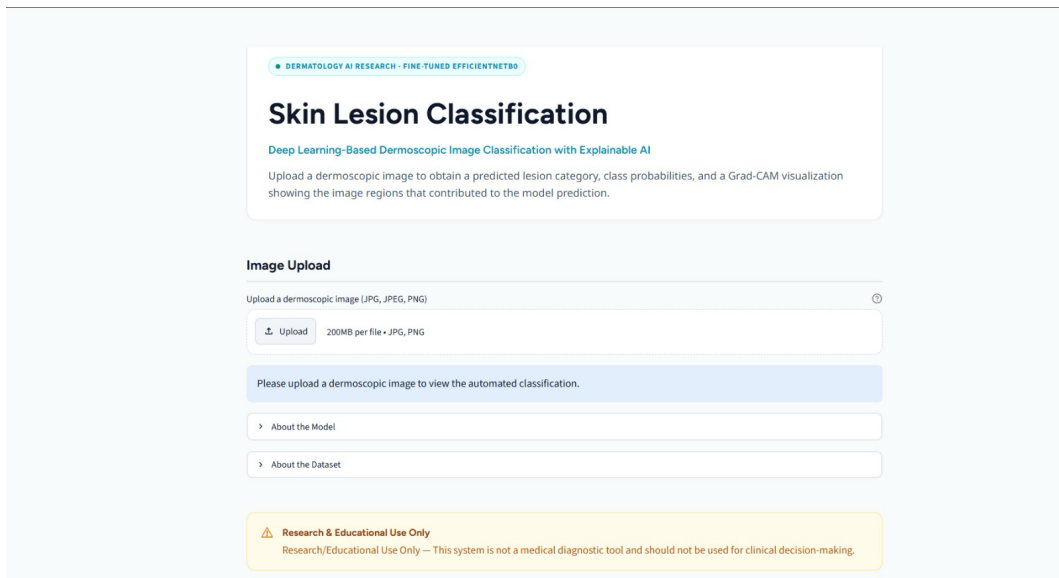
Table 5.5: Comparison with Selected Existing Works

Study	Year	Main Approach	Dataset / Setting	Reported Result	Present Project
Hosny et al. [19]	2020	AlexNet transfer learning	Seven-class lesion classification	Transfer-learning based classification	EfficientNetB0 fine-tuning
Gessert et al. [17]	2020	Multi-resolution EfficientNet ensemble	ISIC 2019	Ensemble-based classification	Single EfficientNetB0
Calderón et al. [15]	2021	BILSK bilinear CNN	HAM10000	93.21% accuracy	75.34% accuracy
Qian et al. [10]	2022	Multi-scale attention + class-specific loss	HAM10000	91.6% accuracy; 97.1% AUC	75.34% accuracy
Manole et al. [6]	2024	EfficientNet-based classification	Dermatological images	EfficientNet-based approach	EfficientNetB0 fine-tuning
Present Project	2026	CNN baseline + EfficientNetB0 transfer learning and fine-tuning + Grad-CAM	11,720 images, eight classes	75.34% accuracy; 76.25% weighted F1	—

The comparison indicates that several published studies report higher accuracy than the present project. These values, however, were obtained under different experimental conditions and often use seven-class HAM10000 or other dataset configurations. Consequently, the values should not be interpreted as directly equivalent benchmarks. The present work focuses on an eight-class dataset and an end-to-end implementation incorporating classification, evaluation, explainability, and deployment.

5.5 Project GUI Snapshots

The developed Streamlit application provides an interactive interface for testing the trained EfficientNetB0 model.

**Figure 5.5:** Main Application Interface

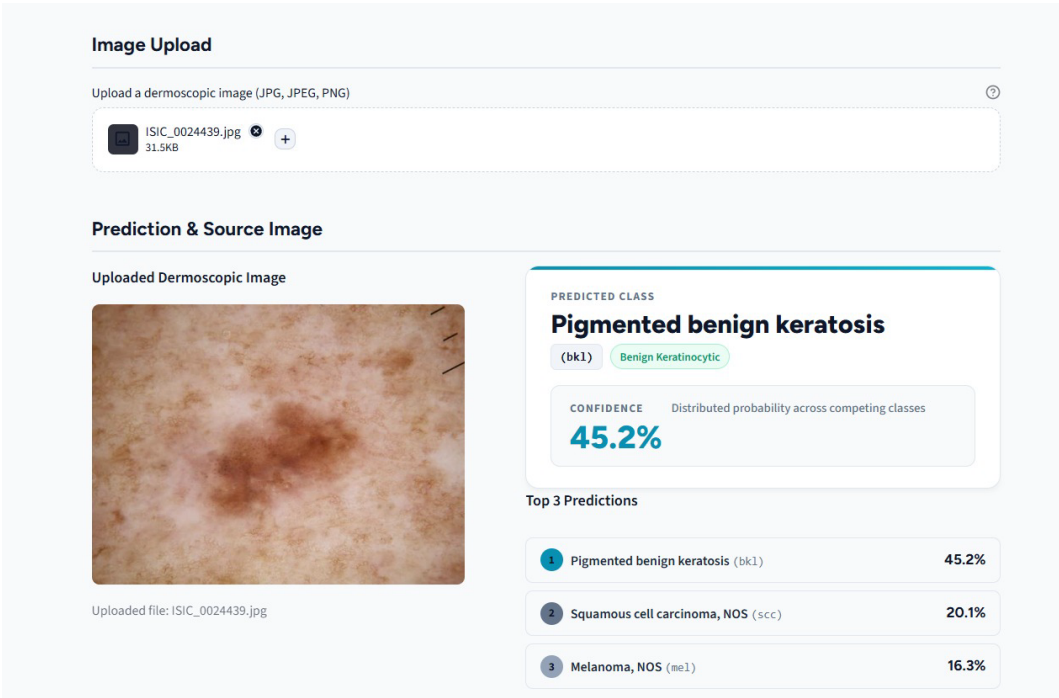


Figure 5.6: Image Upload and Prediction

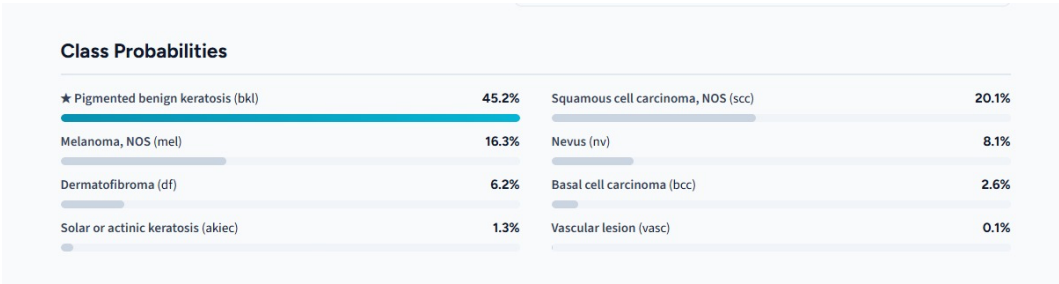


Figure 5.7: Class Probability Distribution

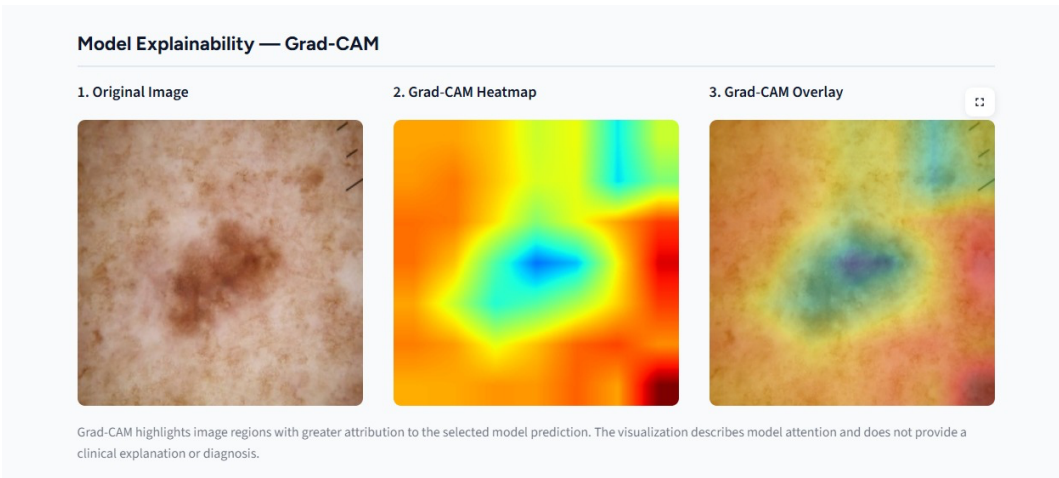


Figure 5.8: Grad-CAM Visualization

Description:

The GUI provides a complete inference workflow in which the user supplies an image, receives the predicted lesion category and confidence, examines the probability distribution, and views the Grad-CAM visualization associated with the prediction.

5.6 Societal Impact of the Project

Automated analysis of dermoscopic images has potential applications in academic research, medical-image analysis, and development of computer-assisted decision-support technologies. The proposed project demonstrates how deep learning can be used to analyze visual patterns in skin lesion images and how model predictions can be accompanied by an explainability visualization.

From an educational perspective, the project provides an end-to-end example of applying convolutional neural networks, transfer learning, model evaluation, explainable AI, and web deployment to a real-world image-classification problem.

The system should not, however, be interpreted as an autonomous medical diagnostic tool. The model operates on image data and does not incorporate the complete clinical context required for medical diagnosis. The project is therefore positioned primarily as an academic and research prototype demonstrating the application of deep learning to skin lesion classification.

5.7 SDG Mapped

The project can be mapped to the following United Nations Sustainable Development Goals (SDGs).

5.7.1 SDG 3 – Good Health and Well-Being

The project contributes to research in medical image analysis and computer-assisted skin lesion assessment. By investigating automated classification of dermoscopic images, it supports technological research related to health and well-being.

5.7.2 SDG 4 – Quality Education

The project provides practical application of deep learning, transfer learning, explainable AI, image classification, and software deployment. It can therefore support technical and research-oriented learning in artificial intelligence and healthcare applications.

5.7.3 SDG 9 – Industry, Innovation and Infrastructure

The project applies modern deep-learning architectures and develops a deployable software interface for an image-classification application. This demonstrates the application of artificial intelligence and digital technologies to an applied research problem.

Table 5.6: SDG Mapping

SDG	Goal	Project Contribution
SDG 3	Good Health and Well-Being	Applies deep learning to skin lesion image analysis
SDG 4	Quality Education	Demonstrates practical AI, deep learning, XAI, and application development
SDG 9	Industry, Innovation and Infrastructure	Develops an AI-based image classification and deployment pipeline

6. Results and Discussion

6.1 Experimental Results

Three model stages were evaluated during the project:

1. Original Custom CNN
2. Stabilized Custom CNN
3. Fine-tuned EfficientNetB0

The models were evaluated on the test dataset using accuracy, precision, recall, F1-score, and class-wise performance.

6.2 Overall Model Performance

Table 6.1: Overall Model Performance

Model	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score	Macro F1-Score
Original Custom CNN	48.84%	69.74%	48.84%	53.20%	—
Stabilized Custom CNN	70.45%	69.44%	70.45%	69.65%	41.59%
Fine-tuned EfficientNetB0	75.34%	77.66%	75.34%	76.25%	52.70%

The results show an increase in overall test accuracy from 48.84% for the original custom CNN to 70.45% for the stabilized custom CNN. The fine-tuned EfficientNetB0 achieved a further increase to 75.34%.

The weighted F1-score also improved from 53.20% for the original model to 69.65% for the stabilized CNN and 76.25% for the fine-tuned EfficientNetB0.

6.3 Class-Wise Performance of EfficientNetB0

The class-wise test results for the final fine-tuned EfficientNetB0 model are shown below.

Table 6.2: Class-Wise Performance of Fine-Tuned EfficientNetB0

Class	Precision	Recall	F1-Score	Support	
akiec	0.38	0.36	0.37	22	
bcc	0.59	0.55	0.57	89	
bk1	0.50	0.63	0.56	191	
df	0.33	0.59	0.42	22	
mel	0.51	0.59	0.55	186	
nv	0.92	0.85	0.88	1106	
scc	0.21	0.21	0.21	33	
vasc	0.65	0.65	0.65	26	

The nv class shows the highest F1-score at 0.88. This is also the class with the largest number of test samples. The vasc class achieves an F1-score of 0.65, while bcc, bk1, and mel achieve F1-scores of 0.57, 0.56, and 0.55, respectively.

The smaller classes, particularly akiec, df, and scc, show lower F1-scores. This indicates that performance varies considerably across the eight categories and that overall accuracy does not fully represent class-level performance.

6.4 Comparison of Model Development Stages

The original custom CNN achieved a test accuracy of 48.84%. Its class-wise results show that several classes were poorly recognized, with zero recall for akiec and df.

After stabilization, the custom CNN achieved 70.45% test accuracy. Performance improved across several classes, although the model continued to show substantial variation between classes. The macro F1-score of 41.59% indicates that performance remained uneven across the eight categories.

The final fine-tuned EfficientNetB0 achieved 75.34% test accuracy and a weighted F1-score of 76.25%. Its macro F1-score increased to 52.70%, indicating improved performance across the class distribution compared with the stabilized CNN.

6.5 Validation and Test Performance

The final EfficientNetB0 model achieved a validation accuracy of approximately 73.54% and a test accuracy of 75.34%.

The corresponding weighted F1-scores were approximately 74.30% on validation data and 76.25% on test data. The relatively similar validation and test results indicate that the final model maintained comparable performance on unseen test data.

However, class-wise performance remains uneven, particularly for classes with fewer samples. Therefore, the model’s overall accuracy should be interpreted together with macro-level and class-specific metrics.

6.6 Confusion Matrix Analysis

The confusion matrix is used to identify which lesion categories are correctly classified and which categories are commonly confused with one another.

The results show strong performance for the nv class, which is reflected by its high recall and F1-score. Other classes exhibit more confusion, particularly the classes with fewer test examples. For example, the model produced some incorrect classifications between bcc and scc, while vasc also had predictions assigned to other categories. Such errors demonstrate that visually similar lesion categories can remain challenging for the classification model.

The confusion matrix therefore provides additional information that is not visible from overall accuracy alone.

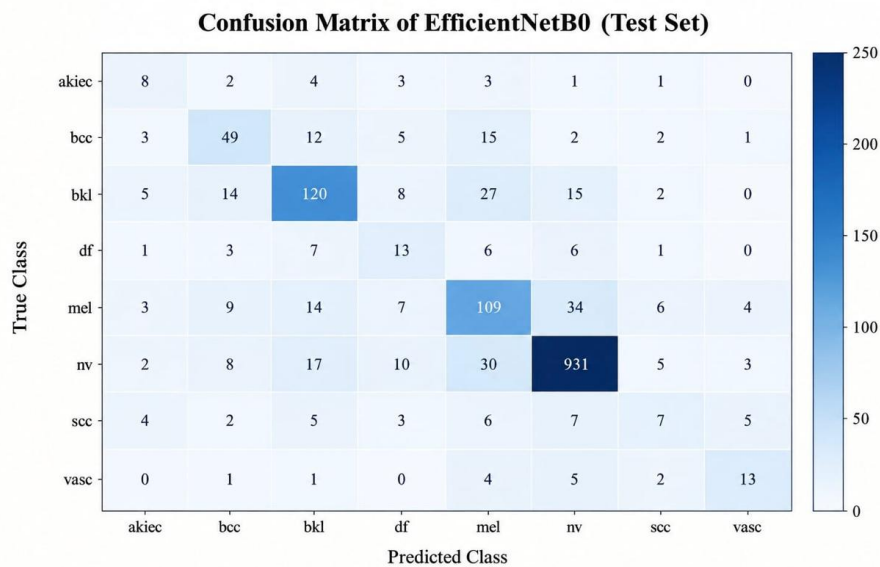


Figure 6.1: Confusion Matrix of Fine-Tuned EfficientNetB0

6.7 Grad-CAM Results

Grad-CAM was generated for representative test images from each of the eight lesion classes. The visualization was used to identify image regions contributing to the model’s predicted class. Examples included both correct and incorrect predictions. For instance, an image with true class bcc was predicted as scc, while an image with true class scc was predicted as bcc. These examples illustrate that the model can encounter difficulty distinguishing visually similar categories.

Grad-CAM provides a visual representation of the regions influencing the model output. It is used in this project as an interpretability aid rather than as a clinical explanation or diagnostic justification.

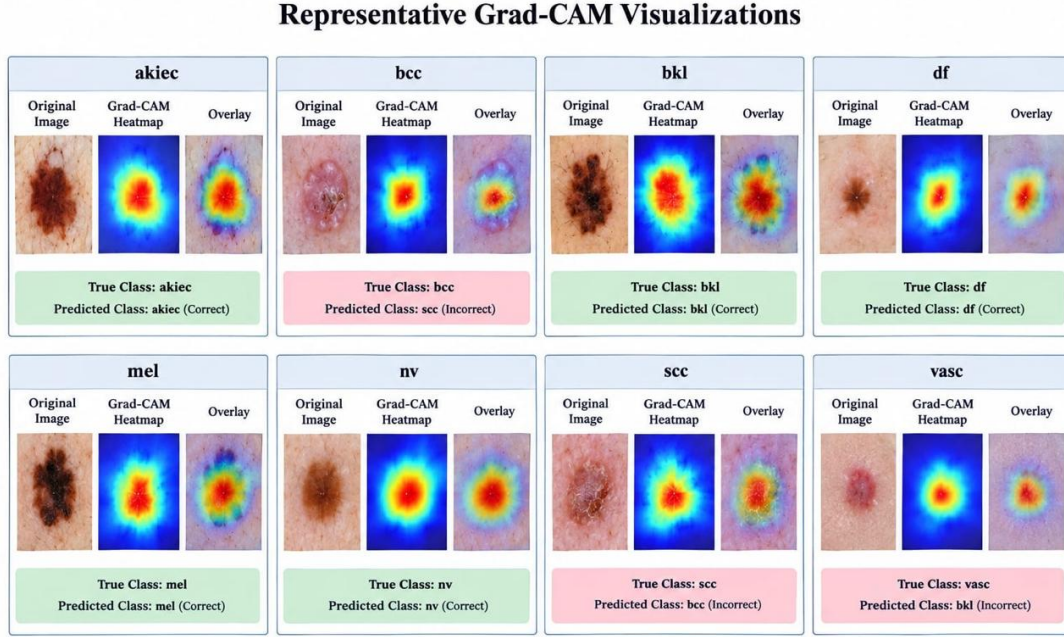


Figure 6.2: Representative Grad-CAM Visualizations

6.8 Discussion

The experimental results demonstrate the effect of progressively improving the model architecture. The custom CNN provides a useful baseline but has difficulty learning robust representations for the complete eight-class problem. Stabilizing the CNN substantially improves overall performance.

The EfficientNetB0 approach provides a further improvement. Transfer learning gives the model access to pretrained visual representations, while fine-tuning allows those representations to adapt to the characteristics of dermoscopic images.

At the same time, the difference between weighted and macro-level performance highlights the effect of class imbalance. The high number of nv samples contributes substantially to the overall metrics, while the smaller classes remain more difficult to classify consistently.

Therefore, the results indicate that overall accuracy should not be considered in isolation. Class-wise precision, recall, F1-score, macro F1-score, and confusion matrix analysis provide a more complete assessment of the model.

6.9 Summary of Results

The final fine-tuned EfficientNetB0 model achieved:

- **Test Accuracy: 75.34%**
- **Weighted Precision: 77.66%**
- **Weighted Recall: 75.34%**
- **Weighted F1-Score: 76.25%**
- **Macro F1-Score: 52.70%**

The results demonstrate that the final model provides improved overall classification performance compared with the custom CNN approaches, while class imbalance and visually similar lesion categories remain important limitations.

7. Conclusion and Future Scope

7.1 Conclusion

This project presents a deep learning-based approach for classifying dermoscopic images into eight predefined skin lesion categories. A custom CNN was initially developed as a baseline, followed by a stabilized CNN and an EfficientNetB0-based transfer learning and fine-tuning approach.

The final fine-tuned EfficientNetB0 model achieved a test accuracy of 75.34% and a weighted F1-score of 76.25%. The results demonstrate an improvement over the custom CNN approaches and show the usefulness of transfer learning for this classification task.

The project also incorporates Grad-CAM to provide visual information about image regions contributing to model predictions. In addition, a Streamlit-based interface was developed to make the trained model accessible through an interactive image-upload workflow.

However, the class-wise results show that performance is not uniform across all eight categories. Classes with fewer samples generally remain more challenging, demonstrating the importance of addressing class imbalance and improving generalization.

The developed system is intended as an academic and research prototype for skin lesion image classification and should not be considered a replacement for professional medical diagnosis or clinical decision-making.

7.2 Limitations

The major limitations of the developed system are:

1. **Class imbalance:** The number of images varies considerably between lesion categories, which can affect classification performance for minority classes.
2. **Class-wise performance variation:** Although the overall test accuracy is 75.34%, performance differs substantially between individual lesion categories.
3. **Visual similarity between classes:** Some lesion categories have similar visual characteristics, resulting in misclassification between certain classes.
4. **Limited clinical scope:** The system performs image-based classification and does not incorporate patient history, physical examination, laboratory information, or other clinical factors.
5. **Interpretability limitations:** Grad-CAM highlights regions associated with the model's prediction, but it does not provide a clinical explanation or establish the presence of a particular medical feature.

7.3 Future Scope

Future development of the project could focus on the following areas:

1. **Improved class balancing:** Additional data collection, targeted augmentation, or class-balancing techniques could be investigated to improve minority-class performance.
2. **Advanced model architectures:** More recent CNN, transformer, or hybrid architectures could be evaluated to determine whether they provide better representation of dermoscopic features.
3. **Larger and more diverse datasets:** Training and evaluation using larger datasets from diverse sources could improve model generalization.
4. **Enhanced explainability:** Additional explainability techniques could be integrated alongside Grad-CAM to provide more comprehensive analysis of model behavior.
5. **Clinical validation:** Before any clinical use, the system would require rigorous external validation, appropriate clinical evaluation, and assessment by qualified medical professionals.
6. **Deployment improvements:** The Streamlit prototype could be extended into a more robust application with improved user interface design, model management, logging, and deployment infrastructure.

7.4 Final Summary

The project successfully demonstrates an end-to-end workflow for eight-class skin lesion classification, covering dataset preparation, CNN-based modelling, transfer learning, fine-tuning, evaluation, explainability, and application development. The final EfficientNetB0 model provides the strongest performance among the implemented approaches, while the remaining class-wise limitations provide clear directions for future research.

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